① Sorted: 124 144 146 147 | 152 154 158 234 $n=8$

1 2 3 4 5 6 7 8

a) $0.25(8+1) = 2.25$ $Q_1 = 144 + 0.75(146-144) = 145.5$

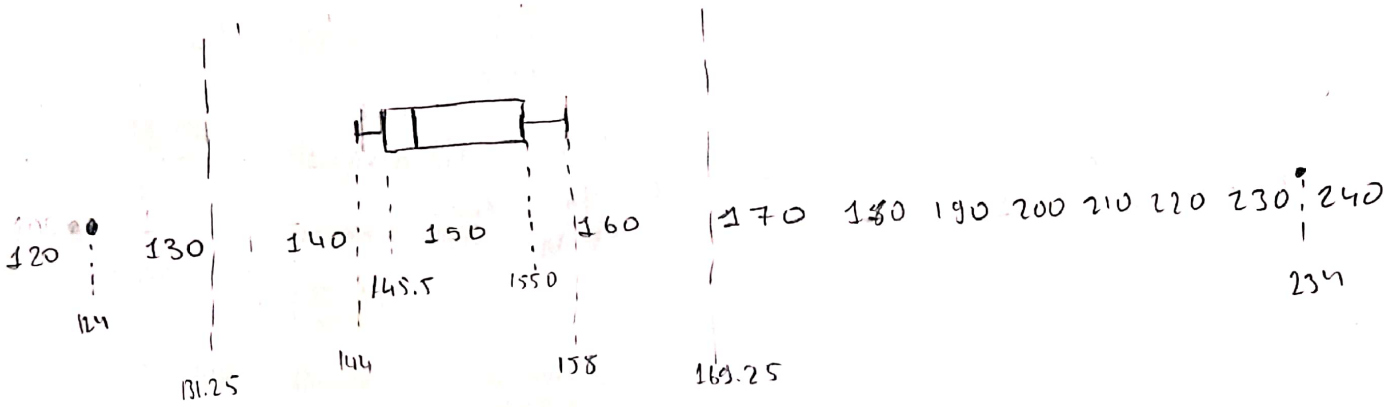
b) $0.75(8+1) = 6.75$ $Q_3 = 154 + 0.25(158-154) = 155.0$

c) $IQR = Q_3 - Q_1 = 155.0 - 145.5 = 9.5$

d) $Q_1 - 1.5\,IQR = 145.5 - \frac{3}{2}(9.5) = 131.25$

e) $Q_3 + 1.5\,IQR = 155 + (1.5 \times 9.5) = 169.25$

f) median = (147+152) / 2 = 149.5



d) Yes, there is some values outside fences. (124 and 234)

$\bar{x} = 157.375$

$s = \frac{\sum (x_i - \bar{x})^2}{7} = 32.60$

z-score: $\frac{124 - 157.375}{32.60} = -1.024$

z-score: $\frac{234 - 157.375}{32.60} = 2.35$

(2) a) $z = \frac{x-\mu}{\sigma}$ $P(x \leq 3.3)$ $z = \frac{3.3-3}{0.3} = 1$

$P(z \leq 1) = 0.8413$ (from the table)

b) $P(x \geq 3.6)$ $z = \frac{3.6-3}{0.3} = 2$ $P(z \geq 2) = 1 - P(z \leq 2)$

$P(z \geq 2) = 1 - 0.9772 = 0.0228$

c) $P(2.4 < x < 3.3)$ $z_1 = \frac{2.4-3}{0.3} = -2$ $z_2 = \frac{3.3-3}{0.3} = 1$ $P(-2 < z < 1) = (1 - P(z \leq 1)) - P(z \leq -2)$

$(1 - 0.8413) - (0.0228) = 0.1359$

d) z-score $= 1.96$ (from the table)

$1.96 = \frac{x-3}{0.3} \Rightarrow 3.588 = x$

Population Variance

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Sample Variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Standard Deviation

σ s $s \approx R/4$

Z-Score

$$\frac{x - \bar{x}}{s}$$

Correlation Coefficient

$$r = \frac{s_{xy}}{s_x s_y}$$

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$s_{xy} = \frac{\sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}}{n-1}$$

Regression Line

$y = a + bx$

$b = r\frac{s_y}{s_x}$ $a = \bar{y} - b\bar{x}$

Probabilities

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Mean $= \mu = \sum x p(x)$

Variance $= \sigma^2 =$

$\sum (x-\mu)^2 p(x)$

Std dev $= \sqrt{\sigma^2}$

Binomial Distribution

$$P(x=k) = C_k^n p^k q^{n-k}$$

Mean $= \mu = np$

Variance $= \sigma^2 = npq$

z-score: $\frac{x-\mu}{\sigma}$ $s = \sqrt{npq}$

Poisson Distribution

$$P(x=k) = \frac{\mu^k e^{-\mu}}{k!}$$ Variance: $\sigma^2 = \mu$

z-score: $\frac{x-\mu}{\sqrt{\mu}}$ Std dev.: $\sqrt{\mu}$

Hypergeometric Dist.

$$P(x=k) = \frac{C_k^M C_{n-k}^{N-M}}{C(N,M)}$$

Mean μ: $n\left(\frac{M}{N}\right)$

Variance σ^2: $n\left(\frac{M}{N}\right)\left(\frac{N-M}{N}\right)\left(\frac{N-n}{N-1}\right)$

Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \doteq \ldots$$

$$z = \frac{x-\mu}{\sigma}$$

$$P(a < x < b) = \int_a^b f(x)\,dx$$